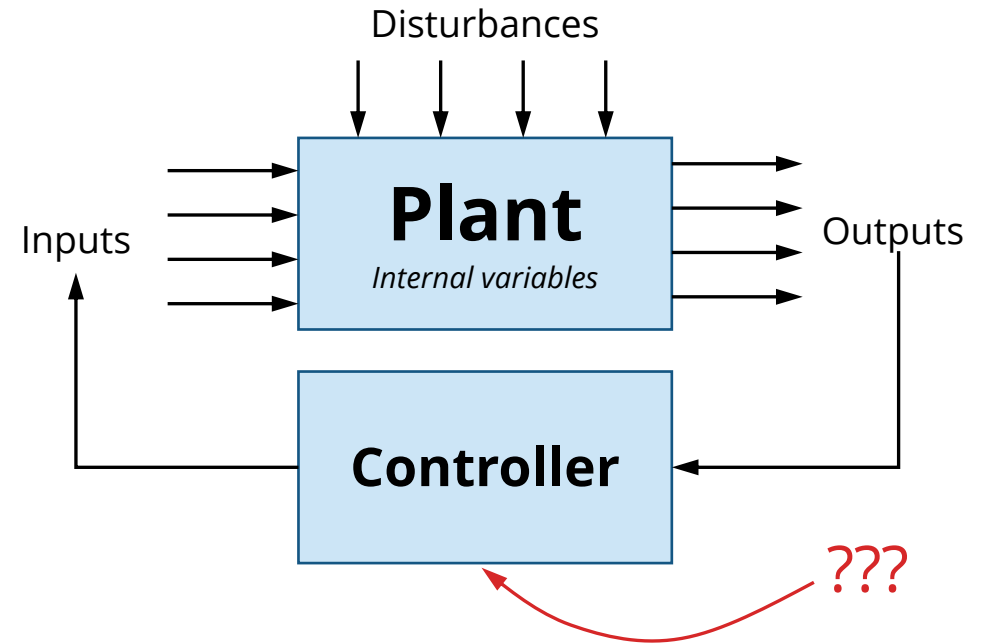




AA 513 Multivariable Control

Lecture 02 // Week 2

Instructor: Prof. Karen Leung

Announcements

- More exercises are due in week 3 (exercises from weeks 1 & 2)
- Clarification on “tutorial”
 - A slight variation to the typical “course project”
 - Recording of you presenting slides on the project details
 - Instead of a technical report, you provide a notebook with text + code + figures.
- TA OH set up Wed 6PM
- Canvas course email corrected
- Extra credit for Ed Discussion participation

Last time

- Introduction to state space representation
 - States (set of variables sufficient to describe the system)
 - Control input (we decide the value of it)
 - Disturbance input (can be used to capture uncertainty in the system)
 - Dynamics (how the states evolve over time)
 - Continuous time vs discrete time
 - Nonlinear vs control affine vs linear
 - Linearizing dynamics
- Coding
 - Basics of defining dynamics and simulating trajectories
 - Euler vs RK4
 - JAX (`vmap`, `jit`, `grad`, `jacobian`, `hessian`)

This week

- Finish up linearization
- Introduction to optimization
- Exercise 2 & 3

Review: Gradients, Jacobian, Hessians

Consider $f: \mathbb{R}^n \rightarrow \mathbb{R}$. $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

$$\nabla f(x)^T = \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right]$$

consider $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$

$$f(x) = \begin{bmatrix} f_1(x) \\ f_2(x) \\ \vdots \\ f_m(x) \end{bmatrix}$$

$$J_f(x) = \nabla f(x)^T = \begin{bmatrix} \text{---} \nabla f_1(x)^T \text{---} \\ \text{---} \nabla f_2(x)^T \text{---} \\ \vdots \\ \text{---} \nabla f_m(x)^T \text{---} \end{bmatrix}$$

Review: Gradients, Jacobian, Hessians

$$\underset{m}{\overset{1}{\left| \right.}} \overset{x \approx x_0}{f(x)} \approx \underset{m}{\overset{1}{\left| \right.}} f(x_0) + \underset{m}{\overset{n}{\left| \right.}} \underset{\substack{\uparrow \\ \text{Jacobian}}}{J_f(x)} \underset{1}{\left| \right.} (x - x_0) + \dots$$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$H_f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \dots & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} \begin{matrix} n \\ n \end{matrix}$$

← symmetric $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$

Introduction to optimization

What is optimization?

- **In words:** The process of finding the *best* decision or design according to a quantitative criterion, subject to constraints that describe what is possible.

- **In math:** $\min_x f(x)$ Measures performance

$\text{s.t. } g(x) \leq 0$ inequality $g(x) = \begin{bmatrix} g_1 \\ g_2 \\ \vdots \\ g_m \end{bmatrix}$
 $\text{subject to } h(x) = 0$ equality

Defines feasibility

Is x the state? No

- **Why?** Provides a systematic way to make decisions. Instead of producing a function or rule to make decisions, we can instead articulate what we want and let an algorithm decide.
- **Connection to controls:** Find best control input to accomplish desired task subject to dynamics and constraints

Applications of optimization

Energy and infrastructure (*Civil/mechanical engineering*)

- *Power grid operation*: optimize generator outputs to meet demand at minimal cost (economic dispatch).
- *Building HVAC systems*: minimize energy while maintaining comfort constraints.

Finance & Business (*Operations research, economics*)

- *Portfolio optimization*: allocate assets to maximize return for a given risk.
- *Supply chain management*: minimize cost of production, transport, and inventory.

Science & Health (*Biology, health sciences, medicine*)

- *Drug dosage optimization*: find schedules that maximize efficacy and minimize side effects.
- *Epidemic control*: determine optimal vaccination or intervention strategies.

Machine Learning

- *Supervised learning*: Given a dataset, find parameters of a model that minimizes prediction accuracy

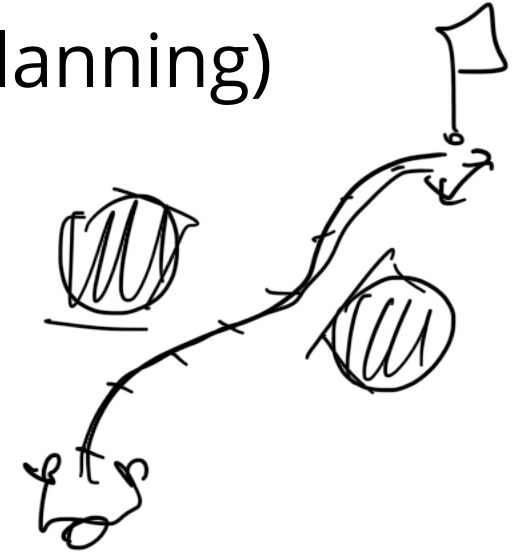
Aerospace applications

Discuss: Objectives and constraints

- Aircraft design and performance
 - Aerodynamics
 - Structures
 - Propulsion
- Spacecraft mission planning
 - Orbit transfer
 - Attitude control
 - Constellation design
 - Mission scheduling
- Operations and logistics
 - Air traffic management
 - Fleet and maintenance scheduling

Aerospace *control* applications

- Flight trajectory and guidance (high-level planning)
 - Trajectory/path planning
 - Landing locations
 - Formation flight
- Control and estimation (navigation)
 - Tracking a trajectory
 - Estimating state



Worst-case analysis

- We can use optimization to find the *worst-case* (pessimization)
- Why?
 - Assess robustness. Still tolerable under worst-case?
 - Falsification / find counter example
 - Collect adversarial data to train model

How to solve optimization problems?

$$\min_x f(x)$$

Measures performance

$$\text{s.t. } g(x) \leq 0$$

Defines feasibility

$$h(x) = 0$$

- From your calculus class: Find stationary points and then ...

$$\min f(x)$$

$$1. f'(x)$$

$$2. f'(x) = 0 \rightarrow \text{find } x^*$$

$$3. f''(x) > 0, < 0 \text{ or } = 0$$



(A)



(B)



(C)

Example: unconstrained

- $f(x) = (x - 3)^2$ (no constraints)

$$f'(x) = 2(x - 3)$$

$$f'(x) = 0 \Rightarrow x - 3 = 0 \quad \therefore x^* = 3$$

$$f''(x) = 2 > 0 \Rightarrow \text{convex } \cup$$

$\Rightarrow \text{minimum!}$

Example: constrained, Lagrangian multipliers

• $f(x) = x^2, \boxed{x \geq 1}$ $g(x) \leq 0$ $\underbrace{1-x}_{g(x)} \leq 0$ in feasible

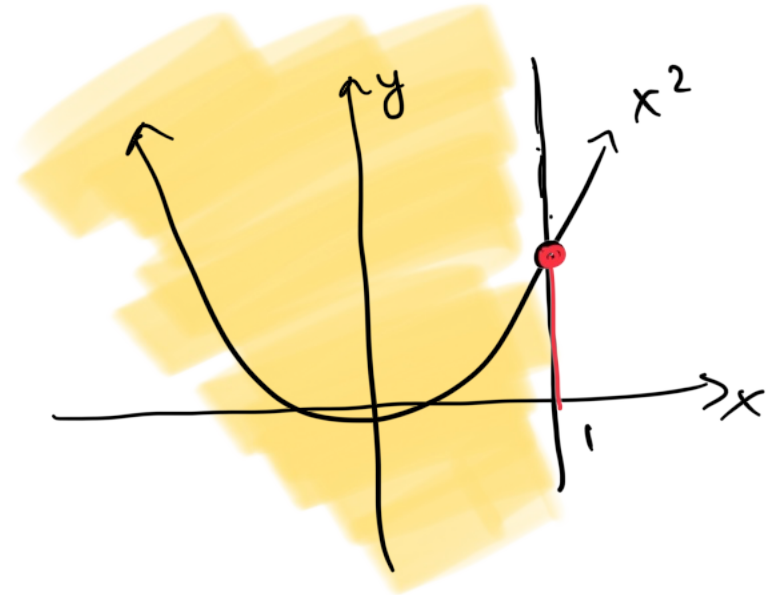
$$\mathcal{L}(x, \lambda) = f(x) + \lambda g(x)$$

$$\mathcal{L}(x, \lambda) = x^2 + \lambda(1-x)$$

$$\frac{\partial \mathcal{L}}{\partial x} = 0 \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 0 \quad \Rightarrow \text{find } x^*, \lambda^*$$

$$\underline{2x - \lambda = 0} \quad \underline{1 - x = 0} \quad \Rightarrow \quad x^* = 1, \lambda^* = 2$$

optimal solution is $x^* = 1$



Necessary and sufficient conditions

- **Necessary** “must have” but by itself is not enough

$$\nabla f(x^*) = 0 \quad (\text{in 1D, } \equiv f'(x) = 0)$$

(More generally, Karush-Kahn-Tucker conditions) (KKT)

- **Sufficient**, enough to show optimality

~~$\nabla f(x^*) = 0$~~

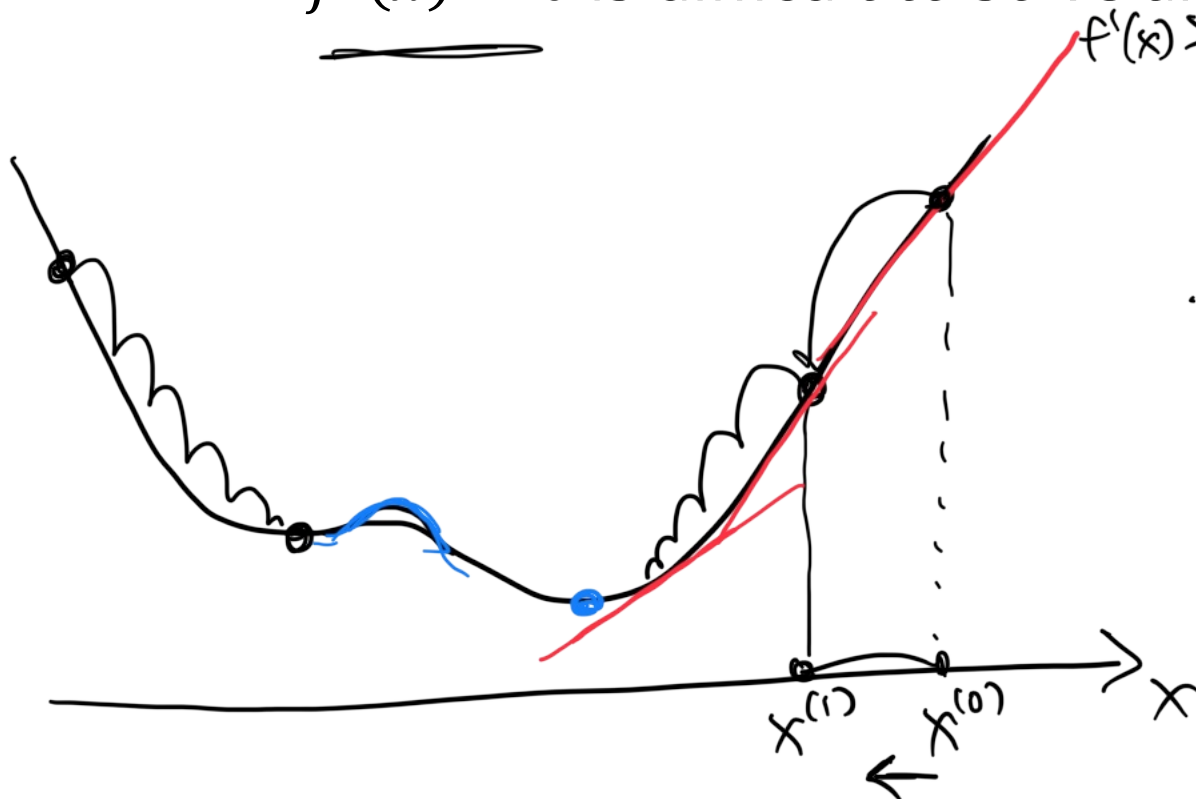
Convexity



(1D setting $f''(x) > 0$)

Gradient descent

- What if finding stationary points is hard?
 - $f'(x) = 0$ is difficult to solve analytically (see exercise 3)



$f'(x) > 0$

$f'(x)$ tells us the direction of greatest ascent

$\therefore -f'(x)$ is the greatest descent

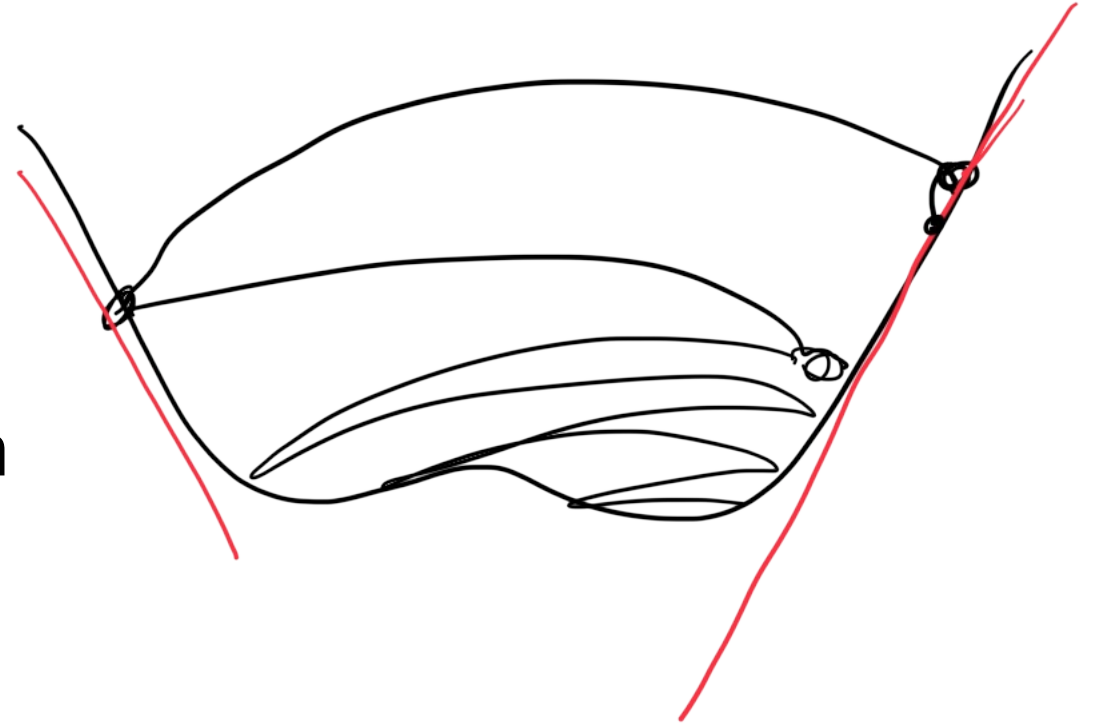
$$x^{(1)} = x^{(0)} - \alpha f'(x^{(0)})$$

α : step size

$$x^{(i+1)} = x^{(i)} - \alpha \nabla f(x^{(i)})$$

Gradient descent: Practical considerations

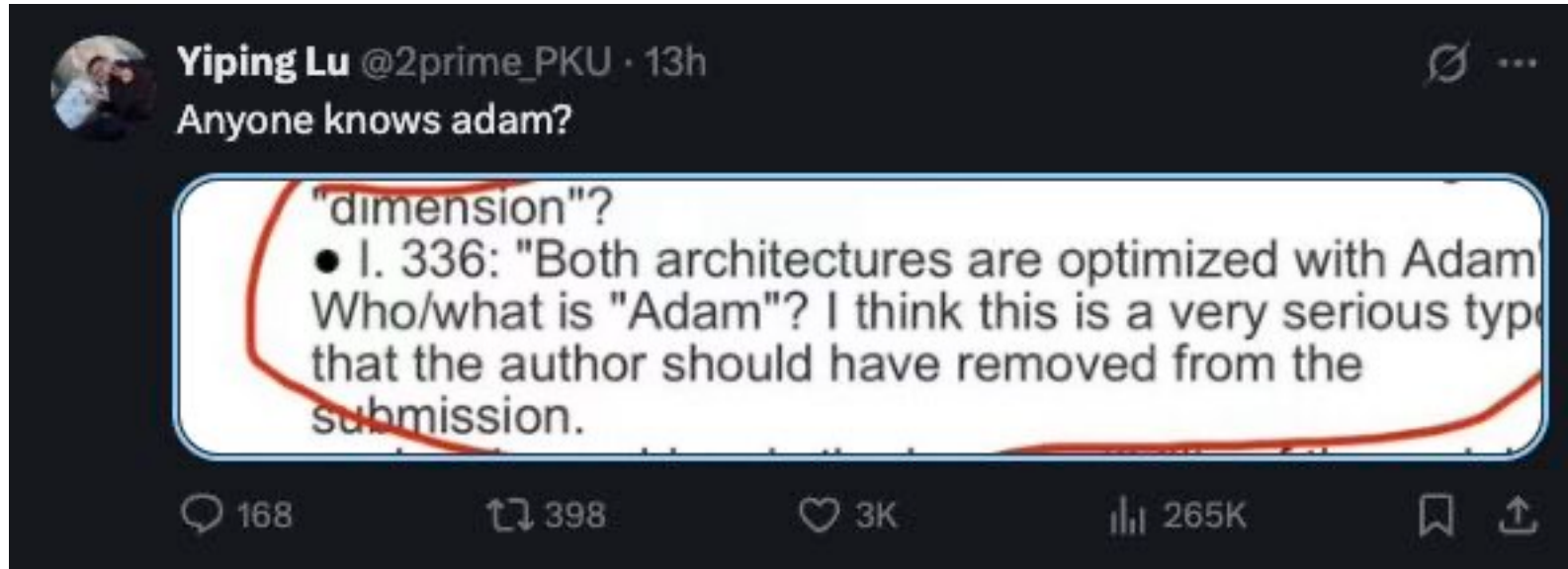
- Step size
- Initial conditions
- Differentiability of the function



More advanced gradient descent algorithms

- Methods to adjust the step size as a function of function landscape
 - **Momentum**: adds inertia, uses previous gradients
 - **Nesterov Accelerated Gradient (NAG)**: Incorporates lookahead
 - **AdaGrad**: uses gradient history
 - **RMSProp**: Improvement on AdaGrad
 - **Adam**: Momentum + RMSProp
 - **Second-order methods**
 - **Line search**: Another optimization problem
- Trade-off between convergence, complexity, computation cost

Side note



- NeurIPS: One of the largest AI/ML research conference
- Some concerns regarding review quality

A whole field on (analytic) optimization theory and tools

- Linear programming (LP)
- Quadratic program (QP) *quadratic cost, linear constraints.*
- Convex optimization (CVX)
- Quadratic constrained quadratic program (QCQP)
- Mixed integer linear program (MILP)
- Nonlinear optimization (~~ML~~) (NL)
- Lagrangian optimization
- Duality
- Quasi-Newton methods
- ...

How to solve optimization problems?

$\min_x f(x)$ Measures performance

s.t. $g(x) \leq 0$
 $h(x) = 0$ Defines feasibility

- Depends on how f, g, h are evaluated
 - Differentiable?
 - Analytic?
 - Cheap or expensive?
 - Known or black box?

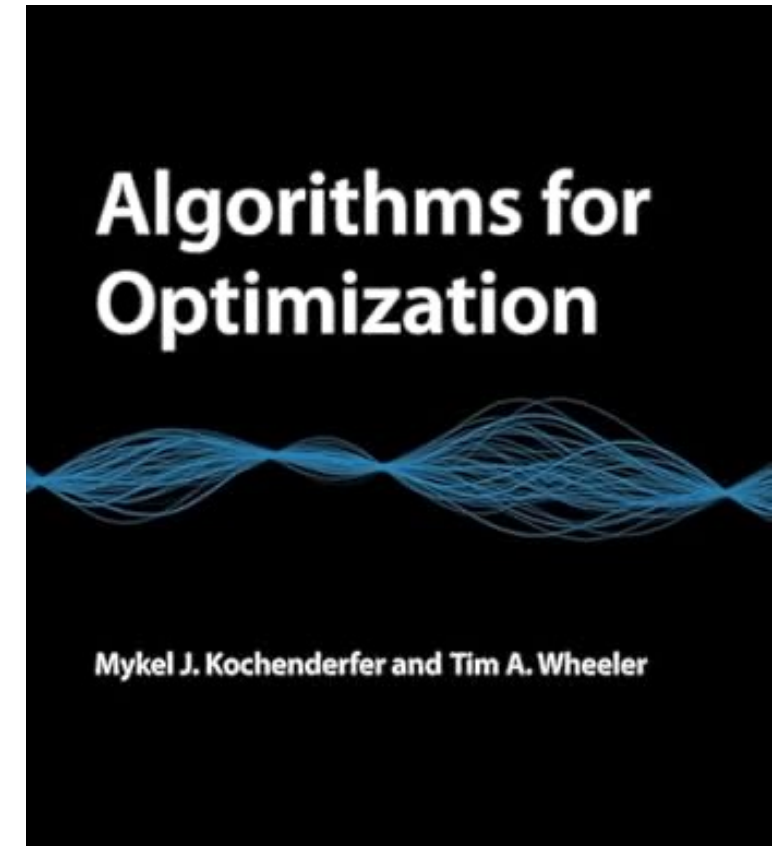
Anti-examples of gradient descent

- Minimize drag on aircraft body
 - Objective = drag
 - Need CFD (black-box) or wind tunnel testing (expensive)
- Choose best wing geometry that minimize drag
 - Straight, swept, delta, elliptical, variable geometry (discrete)
 - Number of wings (discrete)
 - Span, chord, camber (continuous)

(f aircraft design)

So many other optimization algorithms

- Stochastic method
 - Cross-entropy method
 - Simulated annealing
- Population methods
 - Genetic algorithm
 - Cuckoo search
 - Particle swarm optimization
- Surrogate models
- ...



Algorithms for Optimization <https://mykel.kochenderfer.com/textbooks/>

Convex optimization

- A class of optimization problem
 - There are established tools to solve sub-classes of convex optimization reasonably well.
 - (Or at least have hope of finding a reasonably good solution)
- Formulating an optimization problem as convex is the bulk of the challenge.

https://web.stanford.edu/~boyd/cvxbook/bv_cvxslides.pdf

Stephen Boyd.

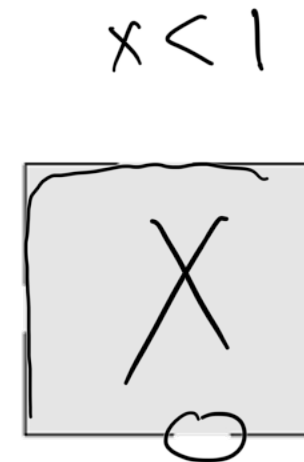
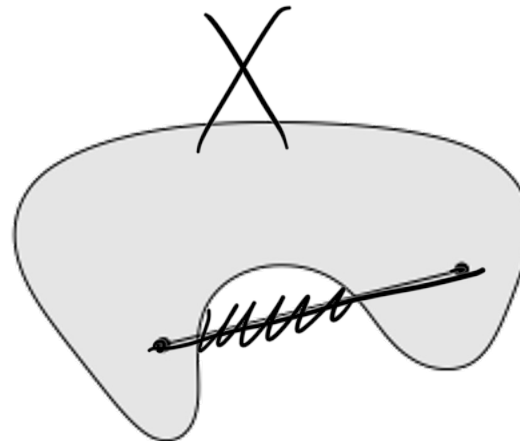
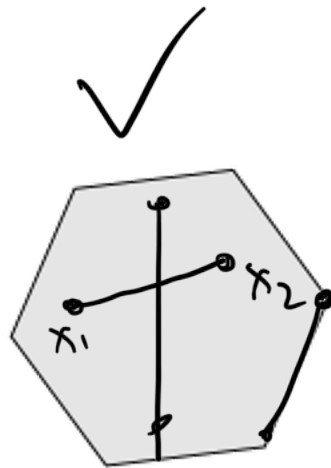
Convex sets

- **Line segment:** line connecting two points

$$x_1, x_2 \quad x = x_1\theta + x_2(1-\theta) \quad , \quad 0 \leq \theta \leq 1$$



- **Convex set:** contains line segment between any two points in the set.

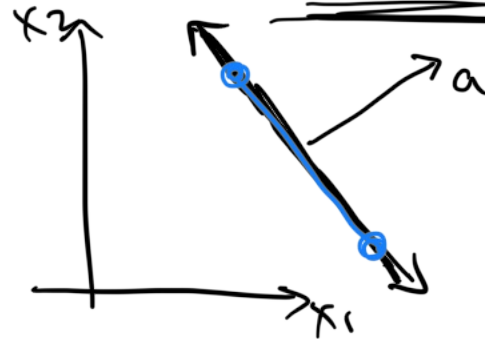


From EE 364A lecture notes

Hyperplane and half spaces

- **Hyperplane:** set of the form $\{x \mid \underline{a^T x = b}, a \neq 0$

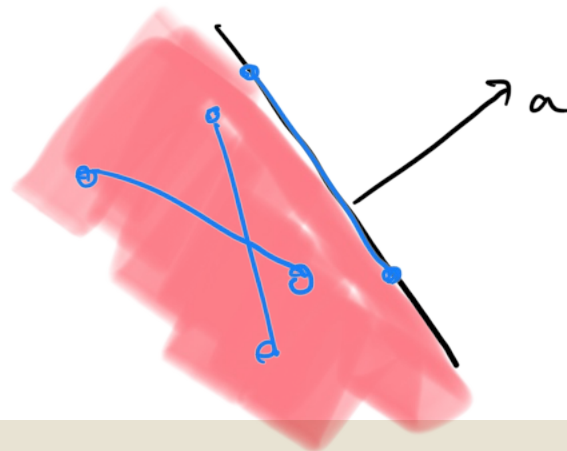
a represents a normal vector



convex?



- **Halfspace:** set of the form $\{x \mid a^T x \leq b\}, a \neq 0$



convex ✓

Question

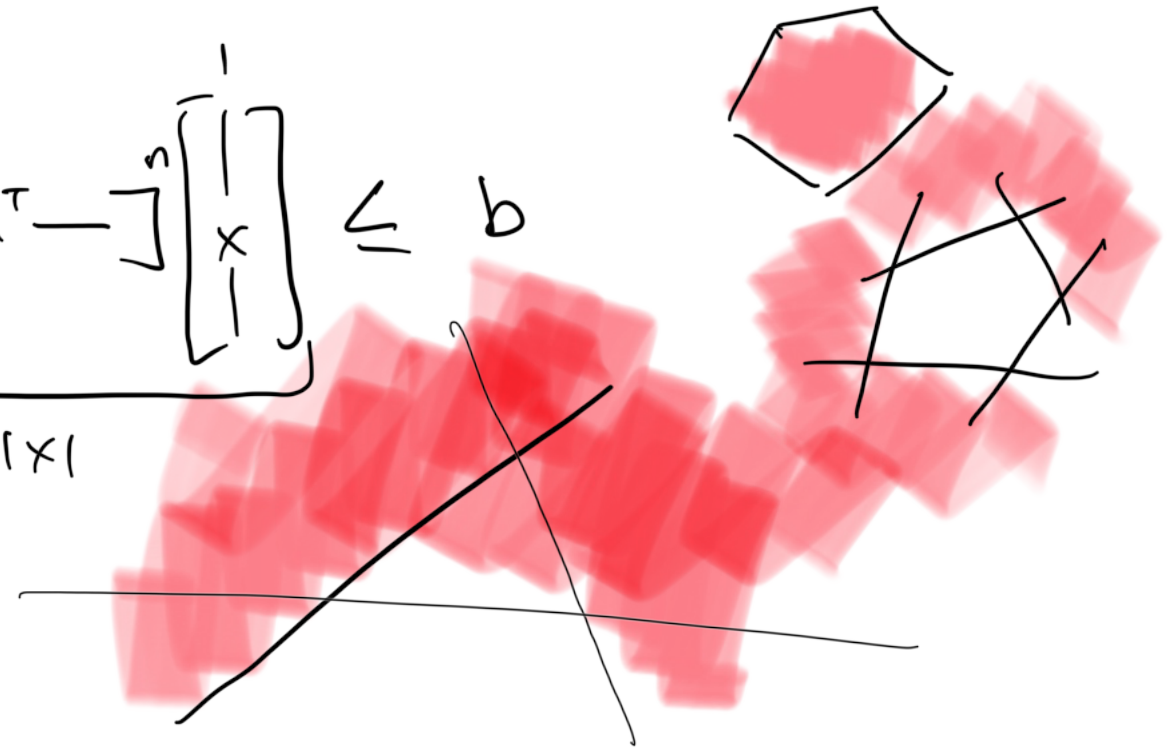
- What does $\{x \mid Ax \leq b\}$, $A \neq 0, A \in \mathbb{R}^{m \times n}$ describe? $b \in \mathbb{R}^m$

A is a matrix

halfspace $a^T x \leq b \Rightarrow \begin{bmatrix} -a^T \end{bmatrix} \begin{bmatrix} x \end{bmatrix} \leq b$

$$\underbrace{\begin{bmatrix} -a_1^T \\ -a_2^T \\ \vdots \\ -a_m^T \end{bmatrix}}_A \begin{bmatrix} x \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

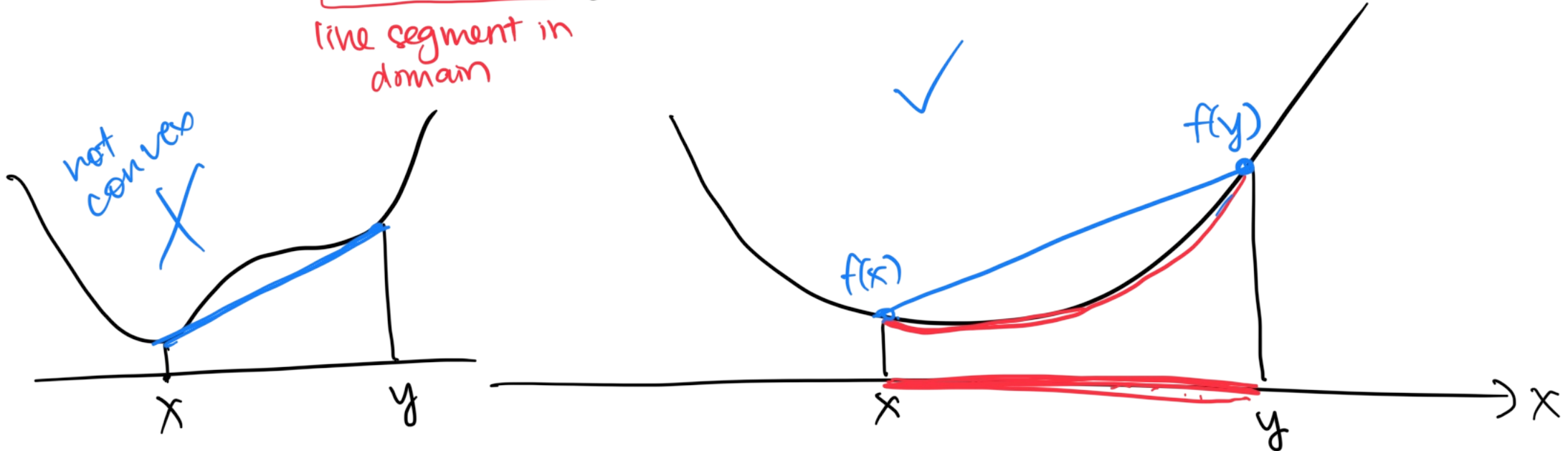
$|x|$



Convex functions

- $f: R^n \rightarrow R$ is convex if $\overset{\text{domain}}{\text{dom } f}$ is a convex set, and for all $x, y \in \text{dom } f, 0 \leq \theta \leq 1 \dots$

$$f(\underbrace{\theta x + (1-\theta)y}_{\text{line segment in domain}}) \leq \theta f(x) + (1-\theta)f(y)$$



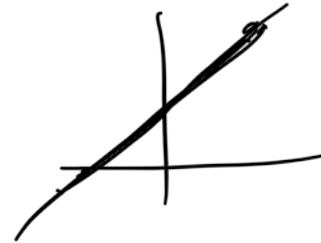
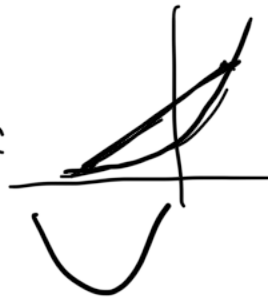
Examples of convex functions

- On $\mathbb{R}$

- Affine $ax+b$

- Exponential e^x

- Quadratic x^2



- On $\mathbb{R}^n$

- Affine $a^T x + b$

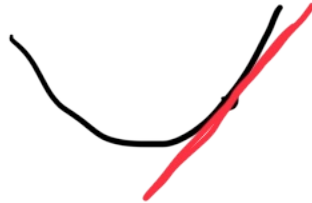
- Norms $\|x\|_p$

- Sum of squares / quadratic

$$\underline{\underline{x^T Q x}}$$

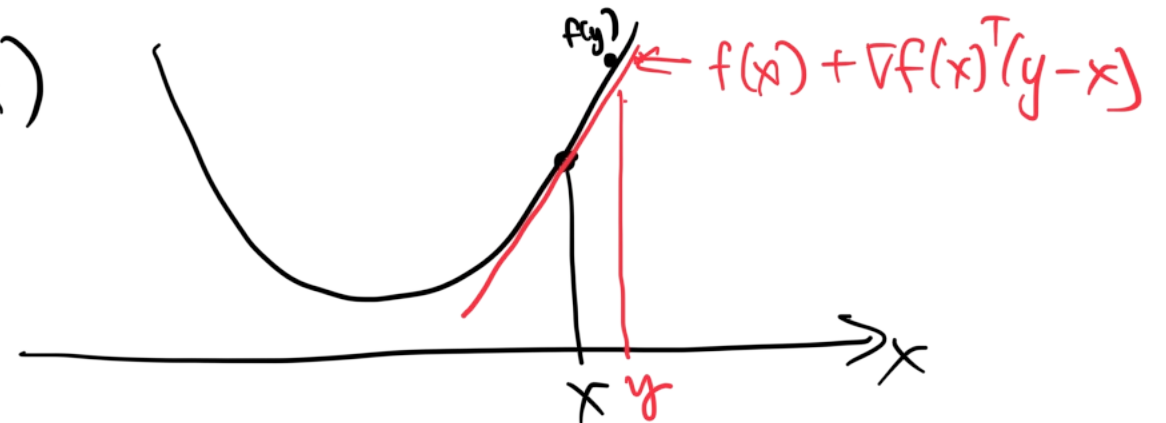
First-order conditions

- Assuming gradient exists
- First order Taylor approximation (tangent) is a global under-estimator



- Differentiable f with convex domain is convex if and only if

$$f(y) \geq f(x) + \nabla f(x)^T (y - x)$$



Second order conditions

- Assuming twice differentiable
 - Twice differentiable f with convex domain is convex if and only if $\nabla^2 f(x) \succcurlyeq 0$
 - $\succcurlyeq$ not $\geq$
 - Positive (semi-)definite
 - Relationship with eigen values?
all $\text{eig}(M) \geq 0$
- $M^T = M$ is PSD iff $x^T M x \geq 0$ any $x \neq 0$

Convex optimization problem

$$\min_x f(x)$$

Convex objective

$$\text{s.t. } g_i(x) \leq 0, \quad i = 1, \dots, m$$

Convex inequality constraints

$$\underbrace{a_i^T x = b_i}_{\text{affine}}, \quad i = 1, \dots, p$$

Affine equality constraints

Quadratic program

suppose we want to
optimize for $\begin{pmatrix} x_0, x_1, \dots, x_{T+1} \\ u_0, u_1, \dots, u_T \end{pmatrix}$ &

$$\min_{\underline{z}} \frac{1}{2} \underline{z}^T P \underline{z} + \underline{q}^T \underline{z} + r$$

quadratic
linear
constant
P PSD

s.t. $G \underline{z} \leq h$
linear inequality cons.

$A \underline{z} = b$
linear equality const.

control effort: $u_t^T u_t$

want x_t to be close to zero. $x_t^T x_t$

$$x_{t+1} = A x_t + B u_t$$

Recurring theme: Turn problems into a QP

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^T Px + q^T x + r \\ \text{s.t.} \quad & Gx \leq h \\ & Ax = b \end{aligned}$$

Exercises

- Gradient descent
- Unconstrained optimization
- Constrained optimization
- CVXPY – python package for solving convex optimization problems